

Foundations of Generative adversarial network

[Generative adversarial network]

$$\min_G \max_D \mathbb{E}[\log D(x)] + \mathbb{E}[\log(1 - D(G(z)))]$$

1.0 Content Block

Knight, Will. "5 Big Predictions for Artificial Intelligence in 2017". MIT Technology Review. Retrieved January 5, 2017. Karras, Tero; Laine, Samuli; Aila, Timo (2018). "A Style-Based Generator Architecture for Generative Adversarial Networks". arXiv:1812.04948 [cs.NE]. This Person Does Not Exist – photorealistic images of people who do not exist, generated by StyleGAN This Cat Does Not Exist Archived March 5, 2019, at the Wayback Machine – photorealistic images of cats who do not exist, generated by StyleGAN Wang, Zhengwei; She, Qi; Ward, Tomas E. (2019). "Generative Adversarial Networks in Computer Vision: A Survey and Taxonomy". arXiv:1906.01529 [cs.LG].

2.0 Content Block

Science Iteratively reconstruct astronomical images Simulate gravitational lensing for dark matter research. Model the distribution of dark matter in a particular direction in space and to predict the gravitational lensing that will occur. Model high energy jet formation and showers through calorimeters of high-energy physics experiments. Approximate bottlenecks in computationally expensive simulations of particle physics experiments. Applications in the context of present and proposed CERN experiments have demonstrated the potential of these methods for accelerating simulation and/or improving simulation fidelity. Reconstruct velocity and scalar fields in turbulent flows. GAN-generated molecules were validated experimentally in mice.

3.0 Content Block

Mathematical The original GAN is defined as the following game: Each probability space $(\Omega, \mu_{\text{ref}})$ defines a GAN game. There are 2 players: generator and discriminator. The generator's strategy set is $\mathcal{P}(\Omega)$, the set of all probability measures μ_G on Ω . The discriminator's strategy set is the set of Markov kernels $\mu_D: \Omega \rightarrow \mathcal{P}[0, 1]$, where $\mathcal{P}[0, 1]$ is the set of probability measures on $[0, 1]$. The GAN game is a zero-sum game, with objective function $L(\mu_G, \mu_D) := \mathbb{E}_{x \sim \mu_{\text{ref}}}[y] - \mathbb{E}_{x \sim \mu_G}[y] + \mathbb{E}_{x \sim \mu_D}[1 - y]$.

4.0 Content Block

CycleGAN CycleGAN is an architecture for performing translations between two domains, such as between photos of horses and photos of zebras, or photos of night cities and photos of day cities.

5.0 Content Block

Equilibrium when discriminator moves first, and generator moves second: $\hat{\mu}_D \in \arg \max_{\mu_D} \min_{\mu_G} L(\mu_G, \mu_D)$, $\hat{\mu}_G \in \arg \min_{\mu_G} L(\mu_G, \hat{\mu}_D)$.

$$\{\mu\}_{D\}},\}$$

6.0 Content Block

For general games, these equilibria do not have to agree, or even to exist. For the original GAN game, these equilibria all exist, and are all equal. However, for more general GAN games, these do not necessarily exist, or agree.

7.0 Content Block

$L_G = E_{x \sim \mu_G} [(D(x) - c)^2]$ where a, b, c are parameters to be chosen. The authors recommended $a = -1, b = 1, c = 0$.

8.0 Content Block

StyleGAN-2 StyleGAN-2 improves upon StyleGAN-1, by using the style latent vector to transform the convolution layer's weights instead, thus solving the "blob" problem. This was updated by the StyleGAN-2-ADA ("ADA" stands for "adaptive"), which uses invertible data augmentation as described above. It also tunes the amount of data augmentation applied by starting at zero, and gradually increasing it until an "overfitting heuristic" reaches a target level, thus the name "adaptive".

9.0 Content Block

InfoGAN In conditional GAN, the generator receives both a noise vector z and a label c , and produces an image $G(z, c)$. The discriminator receives image-label pairs (x, c) , and computes $D(x, c)$. When the training dataset is unlabeled, conditional GAN does not work directly. The idea of InfoGAN is to decree that every latent vector in the latent space can be decomposed as (z, c) : an incompressible noise part z , and an informative label part c , and encourage the generator to comply with the decree, by encouraging it to maximize $I(c, G(z, c))$, the mutual information between c and $G(z, c)$, while making no demands on the mutual information z between $G(z, c)$. Unfortunately, $I(c, G(z, c))$ is intractable in general. The key idea of InfoGAN is Variational Mutual Information Maximization: indirectly maximize it by maximizing a lower bound $I^{\wedge}(G, Q) = E_{z \sim \mu_Z, c \sim \mu_C} [\ln Q(c | G(z, c))]; I(c, G(z, c)) \geq \sup_Q I^{\wedge}(G, Q)$ where Q ranges over all Markov kernels of type $Q: \Omega_Y \rightarrow P(\Omega_C)$.

10.0 Content Block

The discriminator's strategy set is the set of Markov kernels $\mu_D: (\Omega, \mathcal{B}) \rightarrow P([0, 1], \mathcal{B}([0, 1]))$, where $\mathcal{B}([0, 1])$ is the Borel σ -algebra on $[0, 1]$. Since issues of measurability never arise in practice, these will not concern us further.